



NURUL AMELIA

+6285228338362 | na.nurulamelia@gmail.com | <https://www.linkedin.com/in/nramelia2/> | <https://github.com/nramelia2>

Bandung, West Java, Indonesia

Master's graduate in Electrical Engineering with hands-on experience in Data Engineering, Data Analytics, and Artificial Intelligence research within telecommunications and academic environments. Experienced in designing ETL pipelines, managing large-scale operational datasets, developing KPI monitoring dashboards, and transforming complex data into actionable business insights using Python and SQL. Skilled in data warehousing, dimensional data modeling, reporting automation, and business intelligence solutions. Proven ability to bridge industry data engineering practices with research-driven problem solving through telecommunications experience, AI research projects, and six IEEE publications in Computer Vision and Machine Learning.

Technical :

- Programming : Python, SQL, JavaScript
- Data Analyst : Data Cleaning, Data Wrangling, Exploratory Data Analysis (EDA), Data Interpretation, Statistical Analysis, KPI Analysis
- Data Visualization : Power BI, Grafana, Metabase, Microsoft Excel
- Database : MySQL, PostgreSQL, SQL Server
- Data Engineering : ETL/ELT, Data Pipelines, Data Warehousing, Data Modeling (Fact & Dimension Tables)
- Artificial Intelligence : Machine Learning, Deep Learning, Computer Vision
- Additional Tools & Familiarity : Apache Airflow, Apache Spark, SSIS, Pentaho, Talend, QGIS

Work Experiences

ZTE Corporation - Surabaya, Indonesia

PROJECT: Indonesia XL Smart Project MOCN EAST JAVA

Aug 2025 – Feb 2026

Data Engineer

- Designed and maintained ETL pipelines integrating multi-source telecommunications data into a centralized data warehouse, enabling reliable KPI reporting and operational analytics
- Managed and optimized MySQL databases to ensure data integrity, availability, and efficient query performance for large-scale operational datasets
- Built and maintained ETL workflows to support analytics, operational monitoring, and downstream data consumption
- Performed data cleansing, normalization, and transformation to ensure high-quality datasets for analysis and reporting
- Optimized database performance and improved data loading efficiency to support reliable and scalable data operations
- Established and executed backup and recovery procedures to ensure data security, integrity, and availability
- Analyzed network KPI performance and identified trends, irregularities, and potential operational issues through data analysis
- Built KPI monitoring dashboards using Grafana and Metabase, providing faster visibility into network performance trends and operational issues
- Performed data validation, anomaly detection, and ETL testing to improve data quality and prevent inconsistencies in downstream reporting
- Developed automated data workflows using Python and SQL, reducing manual data processing and improving reporting consistency across operational teams
- Compiled 5G Flash Report for 5G Activation using Mongoose Dashboard (ZTE Tools) and Excel VBA
- Designed fact and dimension tables to support scalable analytical reporting and business intelligence initiatives
- Monitored alarm activity and maintained historical records for analysis
- Managed data collection processes related to dump plans and network optimization
- Updated KPI formulas to reflect system upgrades and ensure accurate performance measurement
- Delivered custom queries and modified SQL queries based on tech team requests
- Analyzed network KPIs including throughput, utilization, payload, and active user metrics to identify performance risks and optimization opportunities
- Generated congestion-based cluster rankings and traffic growth analyses to support network evaluation and data-driven optimization planning
- Monitored data patterns and flagged unusual trends for further investigation
- Developed validation scripts and performed ETL testing to ensure data quality, consistency, and pipeline reliability
- Monitored pipeline execution and data flow consistency to ensure stable and reliable reporting operations

Algonova - Work From Home (Part Time)

March 2024 - Aug 2025

Information Technology Tutor

- Taught 200+ students (ages 5–18) in Digital Literacy, Game Design, Python, and AI using adaptive, student-centered methods
- Designed interactive STEM lessons to foster critical thinking and problem-solving
- Maintained 97% student attendance and retention into advanced levels
- Achieved 87–96% teaching quality scores based on reviews and feedback
- Communicated progress and learning plans regularly with parents, providing personalized suggestions and reports

PT Orbit Nasional Edukasi - Jakarta, Indonesia

Sep 2022 - Dec 2022

Junior RPA Developer Intern

- Completed intensive training in UiPath, OutSystems, and TvExtract, focusing on low-code automation and enterprise-grade solutions
- Built a workflow automation prototype for Bank Central Asia (BCA), streamlining the employee payment process via Starpayment
- Delivered 100% of the automation workflow and authored a comprehensive Process Definition Document (PDD) capturing business logic and requirements
- Strengthened business – tech integration through workshops like Turning Dreams into Reality and Business Requirements Analysis, demonstrating structured thinking, precision, and strong collaboration

PT Mitra Inovasi Teknologi - Bandung, Indonesia

Jul 2022 - Aug 2022

RPA Developer Intern

- Underwent structured training in UiPath RPA Developer Fundamentals and Advanced Development
- Built a complete automation project using REFramework, applying process design, error handling, and deployment best practices
- Achieved 100% workflow completion, meeting enterprise-level performance and reliability standards

Physics Laboratory - Bandung, Indonesia

Jul 2020 - Jun 2022

Teaching Assistant

- Assisted Physics I & II lab sessions for Electrical and Industrial Engineering students over 4 academic semesters
- Managed lab setup, including equipment calibration and safety checks, to ensure accurate and secure experiments
- Supported students during experiments, monitored lab activities, and evaluated lab journals and reports
- Developed interactive simulations to facilitate remote lab learning during the COVID-19 pandemic

Research and Organisational Experience

Center of Excellence in Artificial Intelligence for Learning and Optimization

Mar 2023 - Mar 2025

Research Assistant

- Conducted Computer Vision and Deep Learning research that contributed to multiple IEEE publications in AI and remote sensing
- Delivered Machine Learning and Deep Learning training sessions for 50+ participants
- Organized AI-focused knowledge-sharing programs attended by 60+ participants

EXYOLOv8 Project

Nov 2023 - Sep 2024

Artificial Intelligence Engineer

- Developed a custom deep learning model for small object detection in remote sensing imagery using YOLOv8
- Modified the model architecture (backbone, neck, detection head) to improve feature representation and detection precision
- Experimented with various loss functions (GloU, CloU, DloU, SloU, WloU) and identified DloU as the most effective for improving localization accuracy and convergence
- Achieved a +5% increase in mean Average Precision (mAP) compared to baseline YOLOv8 through architectural and loss function optimization
- Conducted structured experimentation and analysis to guide design decisions, demonstrating strong research and implementation skills

Sign Language Translator Using Deep Learning

Apr 2023 - Jul 2023

Artificial Intelligence Engineer

- Developed a real-time Indonesian Sign Language recognition system using YOLOv5 and PyTorch
- Achieved approximately 99.5% classification accuracy through data preprocessing, augmentation, and model optimization
- Collaborated with UI/UX and backend teams to deploy the solution using Flask, Docker, and Google Cloud Platform

Physics Laboratory

Jul 2020 - Jun 2022

Research & Equipment Division

- Prepared and calibrated lab equipment for both offline and online practical sessions
- Scheduled and coordinated routine inventory checks and tool maintenance, assigning responsible assistant teams
- Tracked condition of lab instruments and reported issues for repair or replacement
- Documented inventory data and supported equipment readiness across departments
- Collaborated with team members to maintain a functional, safe, and well-managed lab environment

Mobile Shopping App

Sep 2021 - Jan 2022

UI/UX Designer & React Native Developer

- Designed and implemented a mobile interface using Figma and React Native
- Built backend logic and database structure with full user flow (login, cart, checkout)
- Conducted end-to-end testing and completed 97% of a functional prototype
- Applied responsive design principles and integrated API for product management

Education Level

Telkom University - Bandung, Indonesia

Aug 2023 - Jan 2025

Master of Electrical Engineering, Control and Intelligent System - Cum Laude, 3.91/4.00

- Focus: Computer Vision and Artificial Intelligence
- Thesis: EXYOLOv8 – Advanced Structural Optimization of YOLOv8 for Enhanced Small Object Detection in Remote Sensing Imagery

Telkom University - Bandung, Indonesia

Aug 2019 - Aug 2023

Bachelor of Computer Engineering - Cum Laude, 3.77/4.00

- Capstone Project: Sign Language Translator Using Deep Learning

Achievements & Other Experience

- **Publication** : Published 6 research papers on AI and Computer Vision
- **Achievements** (2024) : Best Paper Award: Recognized for presenting Improved Small Object Detection with YOLOv8 in Remote Sensing Imagery : Architecture Modifications and Loss Function Optimization at the 7th International Conference on Information and Communications Technology (ICOIACT 2024), showcasing innovative research and outstanding contribution to the field

PORTOFOLIO

<https://nurul-amelia.vercel.app/>

DATA VISUALIZATION WEB BASE

https://drive.google.com/file/d/1EcT0RiY47TAqs4eQVxEDM_IrcvGohQxb/view?usp=sharing

DATA VISUALIZATION USING POWER BI

